

Cloud with Azure – Getting Started Plan

(Cloud Fundamentals + Azure Intro & Core Services)

Phase 1: Cloud Fundamentals

Module 1: Cloud Computing Fundamentals

What is Cloud Computing?

- Definition
- Evolution of IT infrastructure
 - Physical Servers
 - Virtualization
 - Private Cloud
 - Public Cloud

Characteristics of Cloud

- On-demand
- Self-service
- Elasticity
- Measured service
- Global reach

Why Cloud?

- Problems with Traditional Data Centers
 - Hardware procurement delays
 - Capacity planning issues
 - High upfront cost
 - Disaster recovery challenges
 - Maintenance overhead
- Benefits
 - Scalability
 - Elasticity
 - High Availability

- Disaster Recovery
- Cost Optimization
- Global Deployment

Module 2: Cloud Economics

- CapEx vs OpEx
 - Traditional: Buy servers, storage, networking equipment
 - Cloud: Rent resources, pay only for usage
- Pay-As-You-Go
 - Examples: Compute billing, Storage billing, Data transfer billing
- Reserved Capacity Concepts (concept only, Azure implementation later)
 - Pay-As-You-Go
 - Reserved Instances
 - Spot Instances

Module 3: Shared Responsibility Model

Very important before touching Azure.

- **IaaS**
 - Customer manages: OS, Applications, Security patches
 - Provider manages: Hardware, Network, Hypervisor
- **PaaS**
 - Customer manages: Application, Data
 - Provider manages: OS, Runtime, Infrastructure
- **SaaS**
 - Customer manages: Users, Data
 - Provider manages everything else

Module 4: Cloud Deployment Models

- Public Cloud — Examples: Azure, AWS, Google Cloud
 - Private Cloud — Examples: Azure Stack, VMware, OpenStack
 - Hybrid Cloud — On-Prem + Cloud (Azure Arc as a preview concept)
 - Multi-Cloud — Azure + AWS
 - Real industry use cases
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Phase 2: Azure Foundations

Before touching services, understand how Azure is organized.

Module 5: Azure Global Infrastructure

Concepts

- Region
- Region Pair
- Availability Zone
- Availability Set (Azure-specific – how it differs from AZ)
- Edge Location / Point of Presence (Azure CDN/Front Door context)

Explain

- Why Azure organizes resources into Resource Groups
- Why South India / Central India regions matter for latency
- Why Availability Zones matter for High Availability
- Availability Set vs Availability Zone – when to use which

Hands-on

- Explore Azure Portal
- Explore Azure Resource Manager (ARM) hierarchy: Management Group → Subscription → Resource Group → Resource
- Change regions and observe service availability
- Explore Azure Cloud Shell (Bash/PowerShell)

Module 6: Azure Account, Subscription & Billing

Topics

- Creating an Azure Account (Microsoft Account / Work or School Account)
- Azure Free Tier
 - Free credits (₹ / \$ amount, validity period)
 - Always-free services
 - 12-months-free services
- Subscriptions
 - What is a Subscription
 - Subscription types: Free Trial, Pay-As-You-Go, Student, Enterprise Agreement (EA), CSP
 - Subscription vs Tenant vs Account – clearing the confusion
- Management Groups (concept overview)
- Resource Groups
 - Purpose of Resource Groups

- Naming conventions
 - Resource Group as a lifecycle boundary
- Azure Cost Management
 - Cost Analysis
 - Budgets and Alerts
 - Avoiding surprise billing (very important for beginners)
- Azure Support Plans (overview: Basic, Developer, Standard)

Hands-on

- Sign up for Azure Free Account
 - Set a spending limit / budget alert
 - Create a Resource Group
 - Explore Cost Management dashboard
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Phase 3: Azure Networking

Relate everything back to Linux Networking fundamentals learned earlier.

Module 8: Virtual Network (VNet) Deep Dive

Networking Refresher

- CIDR notation
- IP Addressing (Private vs Public)
- Routing basics
- NAT concept

Virtual Network (VNet)

- What is a VNet
- Address Space planning (CIDR)
- Region-bound nature of a VNet

Subnets

- Public Subnet
- Private Subnet
- Subnet address ranges within VNet CIDR
- Reserved IP addresses in every Azure subnet (important gotcha for beginners)

Gateways

- Internet connectivity for a VNet (implicit outbound vs explicit)
- NAT Gateway
- Virtual Network Gateway (VPN Gateway) – overview
- ExpressRoute Gateway – overview only (not deep, just awareness)

Routing

- System Routes (default routes Azure creates)
- User Defined Routes (UDR)
- Route Tables

Security Layer

- Network Security Group (NSG)
 - Inbound rules
 - Outbound rules
 - Rule priority
 - Default rules
 - NSG at Subnet level vs NIC level
- Application Security Group (ASG)
 - What problem ASG solves
 - Grouping VMs logically instead of by IP
 - Using ASG inside NSG rules
- NSG vs ASG – how they work together

Firewalls

- Azure Firewall (overview)
 - Stateful firewall as a service
 - Difference from NSG (network-layer filtering vs managed firewall service)
- Basic firewall rule concepts: Application rules, Network rules, NAT rules (conceptual only at this stage)

VNet Peering (introductory concept)

- Connecting two VNets
- Why it matters for future multi-tier architecture

Diagnostics

- Network Watcher (overview)
- NSG Flow Logs (concept, similar to VPC Flow Logs)

Hands-on Labs

- Lab 1: Create a VNet with one public and one private subnet

- Lab 2: Create and attach an NSG to a subnet
 - Lab 3: Create an ASG and use it inside an NSG rule
 - Lab 4: Deploy a NAT Gateway for private subnet outbound access
 - Lab 5: NSG vs Azure Firewall – basic rule comparison experiment
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Phase 4: Azure Compute Basics

Module 9: Virtual Machines (VMs)

Concepts

- VM Images (Marketplace Images, Custom Images)
- VM Sizes / Series (B-series, D-series – just awareness)
- Availability Options at VM creation: Availability Zone vs Availability Set vs None
- Authentication: SSH Key vs Password
- Custom Data / cloud-init (equivalent to EC2 User Data)
- Network Interface Card (NIC)
- Public IP address (Static vs Dynamic)
- Managed Disks
 - OS Disk
 - Data Disk
- VM Pricing Models
 - Pay-As-You-Go
 - Reserved Instances
 - Spot VMs
- VM extensions (brief overview)

Hands-on

- Launch a Linux VM in the public subnet
 - Launch a web server VM (install nginx/apache)
 - Launch a private VM (no public IP) and access it via a jump box / Azure Bastion
 - Attach a data disk to a VM
 - Associate NSG at NIC level and test connectivity
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Phase 5: Azure Storage Basics

Module 10: Azure Storage Account & Blob Storage

This is one of the most important Azure services – foundation for almost everything else.

Topics

- Storage Account
 - Storage account types (General Purpose v2, Blob-only – overview)
 - Redundancy options: LRS, ZRS, GRS, RA-GRS (concept level)
- Blob Storage
 - Containers
 - Blobs (Block, Append, Page – overview)
 - Access Tiers: Hot, Cool, Archive
 - Blob Lifecycle Management Rules
 - Versioning
 - Soft Delete
- Encryption at rest (default, concept only)
- Static Website Hosting via Blob Storage

Hands-on

- Create a Storage Account
- Create a Blob Container
- Upload/download files
- Host a static website on Blob Storage
- Configure a lifecycle rule to move blobs to Cool/Archive tier

Module 11: Azure Files (brief intro)

- Shared file storage over SMB/NFS
- Mounting Azure File Share on a VM
- Comparison with Blob Storage (when to use which)

Hands-on

- Create a File Share
 - Mount it on a Linux VM
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Suggested Sequence for This Plan

1. Cloud Fundamentals
2. Cloud Economics
3. Shared Responsibility Model
4. Cloud Deployment Models

5. Azure Global Infrastructure
6. Azure Account, Subscription & Billing
7. Azure IAM (Entra ID + RBAC)
8. Virtual Network (VNet, Subnets, NSG, ASG, Gateways, Firewall basics)
9. Virtual Machines
10. Storage (Blob + Files)

Flow: Fundamentals → Account/Billing → Identity → Network → Compute → Storage